SAMARTH GUPTA

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ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
2024 - 2026	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	9.533/10
2021	B.Tech. in Electrical Engineering	Indian Institute of Technology, Delhi	9.398/10
2017	CBSE	Millennium Public School, Meerut	95.4%
2015	CBSE	Dewan Public School, Meerut	10/10

SCHOLASTIC ACHIEVEMENTS

- **Department Rank 1:** Topped both semesters of M.Tech. CSE Department. **Only** candidate with CGPA above 9.0 **(2024)**
- **JEE Advanced:** Secured All India Rank of **815 (GE)** in the Joint Entrance Examination-Advanced in first attempt **(2017)**
- **Regional Mathematics Olympiad (RMO): State Rank 3** in Regional Mathematics Olympiad from U.P. Region **(2015)**
- **KVPY Fellowship:** Awarded the Kishore Vaigyanik Protsahan Yojana Fellowship with an All India Rank of **486 (2016)**
- **JEE Main:** Secured All India Rank of **76 (GE)** in the Joint Entrance Examination-Main amongst **1.2M+** candidates **(2017)**
- **Semester Merit Award:** Ranked in **Top 7%** of batch across 4 semesters (First, Fifth, Seventh, Eighth) **(2017, 2019, 2021)**
- **NTSE:** Awarded fellowship for being amongst **Top 1000** in National Talent Search Exam by Ministry of Education **(2015)**
- **Mathematics Olympiad:** Secured **World Rank 5** in Mathematics Olympiad jointly hosted by IEM and SMART **(2018)**
- **National Standard Examination in Physics/Astronomy:** Cleared NSEA and placed in National **Top 1%** in NSEP **(2016)**
- **UPSC ESE Prelims:** Successfully cleared the Engineering Services Examination Prelims conducted by UPSC **(2024)**

WORK EXPERIENCE

- **Software Engineer at Dhani Loans and Services Limited (July 2021 - May 2022)**
  - Contributed to **backend development** of an established high-scale transaction wallet application serving **25M+** users.
  - **Integrated diverse payment gateways and services:** Successfully integrated Billdesk and Razorpay APIs, along with various biller and KYC-related APIs, to enhance transaction processing and customer onboarding.
  - **Implemented automated reconciliation processes:** via cron jobs, ensuring accuracy and reducing manual effort.
  - **Resolved critical production issues:** Proactively analyzed, reproduced, and resolved a variety of production bugs, minimizing downtime and improving system stability, and ensuring uninterrupted service delivery for end-users.

INTERNSHIPS

- **PrepBytes (Remote) (Dec 2019 - Feb 2020)** : *Worked as a problem setter and student mentor on PrepBytes.com*
  - Made problems of varying difficulty on Combinatorics, Graphs, Trees, Arrays, Dynamic Programming, Sorting.

PROJECTS

- **4D Scene Graph Anticipation — Predicting Future Objects, Boxes & Relations** (M.Tech. project, Prof. Parag Singla):
  - Extend **Action Genome** to 4D with temporally linked nodes/edges. Add 3D cues via **DUSt3R/CUT3R/VGGT**.
  - **Tasks:** *4D SGG* (observed frames) and *Anticipation* (k-step future), including objects missing in the last context frame.
  - **Baselines:** Adapt **STTran** for 4D graphs; add **Neural SDE/ODE** temporal head for continuous-time relation.
  - **Proposed:** A **Diffusion Graph Neural Operator (Neural SDE)** that learns smooth global spatio-temporal evolution of objects and relations from π^3 pointmaps, SMPL meshes, and 3D trajectories. Coupled this with a **graph-conditioned VLM module**, injecting 4D graph facts into a LoRA-tuned multimodal LLM. The combined system anticipates future nodes/edges and performs grounded QA even when entities are partially/fully occluded.
- **Neural Turing Machine-style DP Solver :** Modeled classic dynamic programming (DP) updates as a differentiable neural cell, investigating whether neural architectures can learn generalizable algorithmic behavior for combinatorial and even NP-hard problems (Neural Turing Machine perspective), demonstrating learned **DP-like transitions on knapsack**-style tasks.
- **Implemented a Transformer-based model** for image encoding on the CLEVR dataset, inspired by *Attention Is All You Need*, to explore attention mechanisms for visual reasoning tasks and evaluate their effectiveness on visual queries.
- **Course Projects (2024 - 2025) :** Implemented diverse projects including *Havannah game (AI)*, *LaTeX-to-Markdown Converter*, *Contra-style game development*, enhanced xv6 with *Demand Paging* and extended system utilities (*CTRL+F/B* navigation, *CTRL+C* signal handling, *chmod* permissions), *5-Stage SimpleRISC Processor* and *ML/DL* projects
- Built an interactive pathfinding visualizer in TypeScript/React using A* with constraints, deployed via GitHub Pages.

TECHNICAL SKILLS

- **Programming Skills** in C, C++, C#, Java, Python, MATLAB, OCaml, Lustre; **Tools & Frameworks:** Git, Regex, PyTorch

EXTRA CURRICULAR ACTIVITIES

- **Competitive Programming - Highest Ratings:** *Codeforces* - **2112(Master)**, *LeetCode* - **2564**, *CodeChef* - **2059**
  - Secured **All India Rank 66** in ACM-ICPC Amritapuri Regionals Round 2019 and **2nd best team** from IIT Delhi
  - **Problem Setter** for Competitive Programming on Platforms like CodeChef, HackerRank and HackerEarth
- **Aarohan Teaching:** Mentored underprivileged students in Mathematics for JEE, enhancing their conceptual clarity.